

Shadow Effects for Representing Hovering-and-Contact States of a Projected Hand in Spatial Augmented Reality

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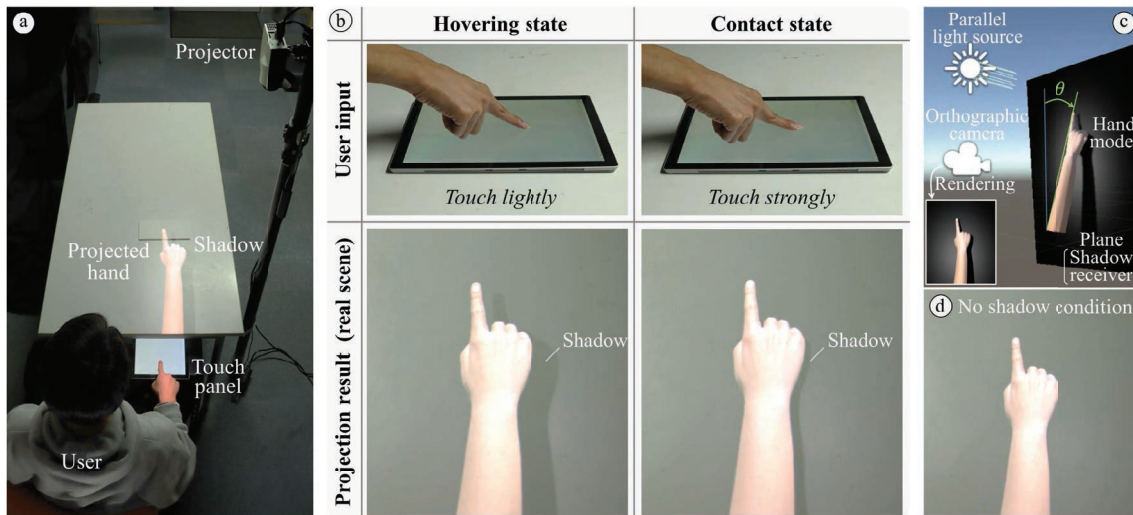


Figure 1: Shadow effects for representing hovering-and-contact states of a projected hand. (a) Overview of the system. (b) Visual appearance of the projected hand based on user touch input. (c) Process for generating projection images. (d) Projected hand under the no shadow condition corresponding to the conventional method.

ABSTRACT

By projecting a user-controllable computer graphics (CG) hand into real-world space, users can interact with out-of-reach objects by pointing or touching them using the projected CG hand (referred to as the projected hand). However, due to the nature of projection, the projected hand cannot appear to hover in mid-air. This study proposes a method that represents a shadow of the projected hand and dynamically alters its appearance based on user input. This allows users to perceive whether the projected hand is hovering or in contact with an object. A user study suggests that the proposed shadow effect effectively enhances the perception of the projected hand's hovering state.

Index Terms: Spatial augmented reality, human augmentation, projected hand, virtual hand illusion.

1 INTRODUCTION

Human augmentation refers to technologies that enhance human physical and sensory capabilities, enabling actions that are impossible with the natural body [7, 14, 13]. Various approaches to human augmentation have been explored, such as wearable robotic arms and powered suits. Among these is ExtendedHand, which utilizes projection-based augmented reality to visually extend the

reach of a user's hand [10, 22]. This interface senses the user's hand movements through devices such as a camera [10] or a touch panel [22], amplifies those movements, and reflects them in the motion of a computer graphics (CG) hand, which is then projected into real space using a projector. This paper will refer to the projected CG hand as the "projected hand." Due to the nature of projection, the projected hand can be displayed over a wide area in real space. This allows users to perform actions such as pointing to or touching objects that are otherwise beyond their reach. However, while the actual hand can make contact with or remain separate from physical objects, the projected hand (including its arm) inevitably appears on the surface of these objects due to the nature of projection. As a result, users often perceive the projected hand as being in contact with the physical objects, making it difficult to recognize it as hovering above the surface.

This paper proposes a method to simulate shadows for a touch-based ExtendedHand [22] and dynamically alter their appearance based on user touch inputs. Additionally, it reports on a user study conducted to investigate whether the proposed method allows users to perceive whether the projected hand is hovering or in contact with an object.

2 PROPOSED METHOD

We represent the hovering and contact states of the projected hand using a hovering level $h \in [0, 1]$, where $h = 0$ indicates full contact with the object's surface, and $h = 1$ represents maximum hovering. Section 2.1 explains the mapping from a user's touch input to the hovering level h , while Section 2.2 describes the shadow representation and the method for generating its appearance based on the hovering level h .

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2.1 Expression of Hovering and Contact States through Touch Input

As discussed in Section 1, this study focuses on the touch-based ExtendedHand interface [22]. While a user operates the projected hand, their hand is always in contact with the touch panel. To enable the user to express the hovering, contact, and intermediate states of the projected hand in this situation, this study utilizes the touch area. Typically, the touch area increases in proportion to the pressing force.

Based on this, our approach expresses a hovering state when the user lightly touches the touch panel and a contact state when the user applies stronger pressure. Specifically, thresholds A_{min} for light pressing and A_{max} for strong pressing are individually determined for each user. If the touch area A is less than A_{min} (including 0, which indicates the user's hand has left the touch panel), the hovering level h is set to 1. If A is greater than A_{max} , h is set to 0. For A between A_{min} and A_{max} , h is linearly interpolated.

2.2 Shadow Effect and its Appearance Control

Shadows are frequently used to represent the hovering of virtual objects [4, 9, 11, 19]. In this study, we follow this convention by employing shadows to convey the hovering sensation of the projected hand. Since negative light, which darkens illuminated areas, does not exist in the real world, we adopt the approach proposed in [9]. Specifically, as shown in Fig. 1(c), the area surrounding the projected hand is brightened, and this brightened region is used to create the shadow effect.

The shadow can be easily rendered using game engines such as Unity, provided the light source, hand model, and shadow receiver plane are properly configured. Although the appearance, location, and consistency of shadows with real-world lighting can influence users [19, 8], as a first step, we configured the light source, hand model, and shadow receiver plane so that the shadow appears to the lower right of the projected hand (see Fig. 1(c)).

To change the appearance of the shadow based on the hovering level h , the pitch angle θ of the hand model (Fig. 1(c)) is defined as:

$$\theta(h) = (1 - h)\theta_{max} \quad (1)$$

where θ_{max} is the pitch angle of the hand model the fingertip makes contact with the shadow receiver plane. This configuration ensures that when $h = 1$ (hovering state), a shadow with a constant width appears on the right side of the projected hand, along the hand and arm. As h decreases, the shadow narrows toward the fingertip, representing a state where the hand approaches the ground while the arm's root (elbow) remains at the same height (Fig. 1(b)).

3 USER STUDY

We conducted a user study to evaluate whether the proposed method enables users to perceive the projected hand as either hovering or in contact with an object.

This experiment was approved by the Ethics Committee of Osaka University (No. R2-28), and all participants signed informed consent forms before participating.

3.1 Experimental Design

3.1.1 Participants

The experiment involved 24 participants (16 males, 23 right-handed, aged 22.5 ± 1.47 years) who were unaware of the study's objectives.

3.1.2 Independent variables

As independent variables, this experiment included two factors: the **Shadow Effect** and **Synchronization**.

The Shadow Effect factor had two conditions: **w/ Shadow** (Fig. 1(b)), where the proposed shadow effect was applied, and **w/o**

Table 1: **Questionnaire**, consisting of seven statements divided into five different categories.

Category	Questionnaire
Hovering	Q1: I felt the projected hand was hovering over the table
Contact	Q2: I felt the projected hand was in contact with the board
Transition agency	Q3: I felt as if the hovering and contact of the projected hand switched according to my actions
Ownership	Q4: I felt as if I was looking at my own hand Q5: I felt as if the projected hand was my hand
Agency	Q6: The projected hand moved just like I wanted it to, as if it was obeying my will Q7: I felt as if I was controlling the movements of the projected hand

Shadow (Fig. 1(d)), where it was not. In the w/ Shadow condition, under the maximum hovering state, the width of the shadow appearing on the right side of the projected hand was approximately 30 mm (see the bottom left of Fig. 1(b)).

The Synchronization factor also had two conditions: **Syn**, where the participant's hand movements controlled the projected hand, and **Asyn**, where pre-recorded hand movements from the experimenter were replayed instead. This factor is frequently used as a control condition in studies on rubber or virtual hand illusions [3, 12], and was also adopted as a control condition in this experiment.

3.1.3 Dependent variables

To evaluate participants' perception of the projected hand as hovering (Hovering) or in contact with objects (Contact), as well as their perception of controlling these states (Transition Agency), we prepared the questionnaire items Q1–Q3 shown in Table 1.

Additionally, we measured the change in participants' proprioceptive sense of their fingers in the gravitational direction before and after using the projected hand (Proprioceptive drift), as this is commonly employed in studies related to rubber or virtual hand illusions [2, 3, 21, 12]. When there is a spatial gap between the real hand and the virtual hand, it is known that the proprioceptive sense of the user's real fingers shifts toward the virtual hand when a sense of ownership is induced. We hypothesized that if the shadow effect led participants to perceive the projected hand as being above the projection surface, the magnitude of this proprioceptive shift would increase; thus, we adopted this measure.

Furthermore, to gain a deeper understanding of the experimental results, we included questionnaire items addressing Ownership and Agency [3] (Q4–Q7 in Table 1), which are commonly investigated in these fields.

3.2 Experimental setup

Figure 2 shows the experimental environment and setup. Participants sat in a chair and operated the projected hand, displayed on a white table surface using a projector (Optoma, ML1050ST+), through a touch panel (Microsoft, Surface Pro 4) positioned 200 mm below the table surface. Additionally, participants wore noise-canceling headphones (Sony, WH-1000XM3) to receive beep sound feedback, as described in Section 3.3. A white paper board (Koyo Sangyo, goo panel; 100 mm × 200 mm × 5 mm) served as the object for touch with the projected hand and was positioned 275 mm from the participants' side of the table edge.

The ceiling lighting in the room was installed to the front left of the participants, ensuring consistency with the shadow effect that appeared on the lower right of the projected hand. The brightness of the ceiling light was adjusted to enable clear visibility of both the projected hand and its shadow. Specifically, the brightness in

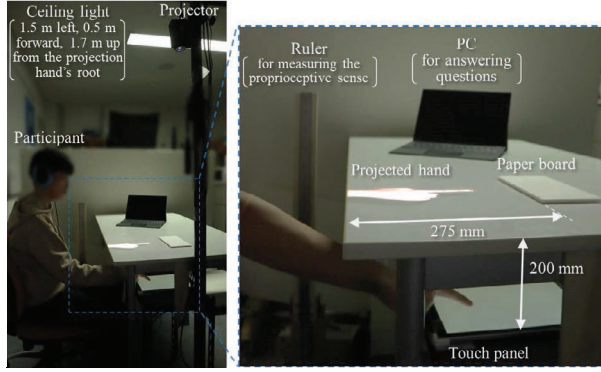


Figure 2: Experimental setup.

the projected hand area was 350 lx, the shadow area was 80 lx, and the slightly brightened area near the shadow, due to projection, was approximately 110 lx. In fact, in a free interview conducted after the experiment, all participants, except for one who did not notice the shadow on the projected hand, reported no discomfort with the direction of the shadow's appearance and did not notice that the area around the projected hand was slightly brighter.

The experimental program was developed using Unity 2021 and ran on a desktop PC (CPU: Intel, Core i7-13700, RAM: 32GB, GPU: NVIDIA, GeForce RTX 4080). The hand model was based on Leap Motion's RiggedSaltFullHands and SkinLightSaltModel-Full. To focus the evaluation solely on the effects of the shadow, the appearance and behavior of the projected hand were designed to closely mimic the participants' real hands. Specifically, the size of the projected hand was adjusted to match the actual hand size of each participant. Additionally, while ExtendedHand normally amplifies the user's hand movements and reflects this amplification in the movement of the projected hand, the amplification rate was set to 1x in this experiment. The projection surface and the touch panel were pre-aligned so that the index finger of the projected hand consistently appeared directly above the touch point of the participant's real hand. The movement range of the projected hand during the experiment was entirely within the touch panel's area. The delay between the participant's hand movement and the projected hand's response was 150 ms.

To measure proprioceptive drift, a ruler was placed on the left side of the participants. The ruler was attached to a linear slider, allowing for vertical adjustments along the direction of gravity. To prevent participants from memorizing and reporting specific ruler values during proprioceptive measurements, the offset position of the ruler was varied for each trial.

3.3 Experimental Procedure

3.3.1 Preparation

Participants were given an explanation of the experiment and signed an informed consent form. The experimenter recorded the participants' age, dominant hand, and hand size. Next, participants were asked to lightly and strongly trace the touch panel to determine the lower threshold A_{min} and upper threshold A_{max} for distinguishing between the two levels of tracing.

Subsequently, participants practiced the main experimental task (Step 2. Tracing Task) under the w/o Shadow and Sync condition. In this task, participants alternately traced the table surface and the paper board every 5 seconds. A beep sound (a 500 Hz sine wave lasting 100 ms) played through headphones indicated the 5-second intervals. Participants were instructed to operate the projected hand with their index finger, tracing the table surface lightly and the pa-

per board strongly. The experimenter monitored participants' input during the task. Specifically, the experimenter ensured that while the projected hand was tracing the table surface, the hovering level h was 0.8 or higher for at least 80% of the time, and while tracing the paper board, h was 0.2 or lower for at least 80% of the time. If these criteria were not met, the experimenter provided verbal feedback. Similarly, verbal feedback was given if the participant moved the projected hand either too slowly (below 100 mm/s) or too quickly (above 500 mm/s). Participants continued practicing until they could consistently perform the required operations.

3.3.2 Main experiment

After practice, participants performed the following steps for each of the four conditions (Shadow Effect $\times$ Synchronization). The order of the four conditions was counterbalanced across participants. Specifically, there were $4! = 24$ possible orders, and each of the 24 participants was assigned a unique order.

Step 1. Pre-Task Proprioceptive Measurement: The system displayed a red dot with a diameter of 10 mm at a point 200 mm away from the edge of the table (hereafter referred to as the home position). Participants were instructed to align the projected index fingertip of their hand with the red dot while lightly touching the touch panel. Once the projected hand reached the designated position, it disappeared from the display. Participants then reported, using a ruler, the perceived vertical position of the tip of their right index fingertip, and the system recorded the height from the tabletop based on the reported value.

Step 2. Tracing Task: To familiarize themselves with the projected hand for each condition, participants performed the alternating tracing task described above for three minutes. Under the Sync and w/ Shadow condition, when participants operated the projected hand as instructed, it hovered above the table surface and made contact with the paper board. In the Async condition, the projected hand moved according to pre-recorded input from the experimenter instead of the participant's touch input. Verbal feedback was provided if participants deviated from the required operations.

Step 3. Post-Task Proprioceptive Measurement: Immediately after completing the tracing task, a red dot was displayed at the home position. Participants followed the same procedure as in Step 1 to report the ruler value, and the system recorded the height from the tabletop based on the reported value.

Step 4. Subjective Questionnaire: Participants used a PC placed on the table (see Fig. 2) to answer the questionnaire items listed in Table 1, using a 7-point Likert scale. The order of the questions was randomized for each trial.

3.4 Results

Figure 3 shows the results of the scores for each category in Table 1 and the proprioceptive drift. The proprioceptive drift was calculated by subtracting the pre-task proprioceptive measurement from the post-task measurement, with positive values indicating that the proprioceptive sense shifted closer to the projected hand after the task.

The scores for each category in Table 1 were analyzed using a non-parametric approach, as they were measured on a Likert scale. Specifically, Aligned Rank Transform (ART) ANOVA [24] was employed for a two-way within-subjects factorial analysis, with Shadow Effect and Synchronization as factors.

For Hovering, no significant difference was observed in the interaction effect ($F(1, 23) = 0.04, p = 0.843, \eta_p^2 = 0.002$). However, significant differences were found in the main effects of both Shadow Effect ($F(1, 23) = 26.95, p < 0.001, \eta_p^2 = 0.540$) and Synchronization ($F(1, 23) = 6.29, p = 0.020, \eta_p^2 = 0.215$).

For Contact, no significant differences were observed in the interaction effect ($F(1, 23) = 0.30, p = 0.592, \eta_p^2 = 0.013$) or the main effect of Shadow Effect ($F(1, 23) = 1.99, p = 0.172, \eta_p^2 =$

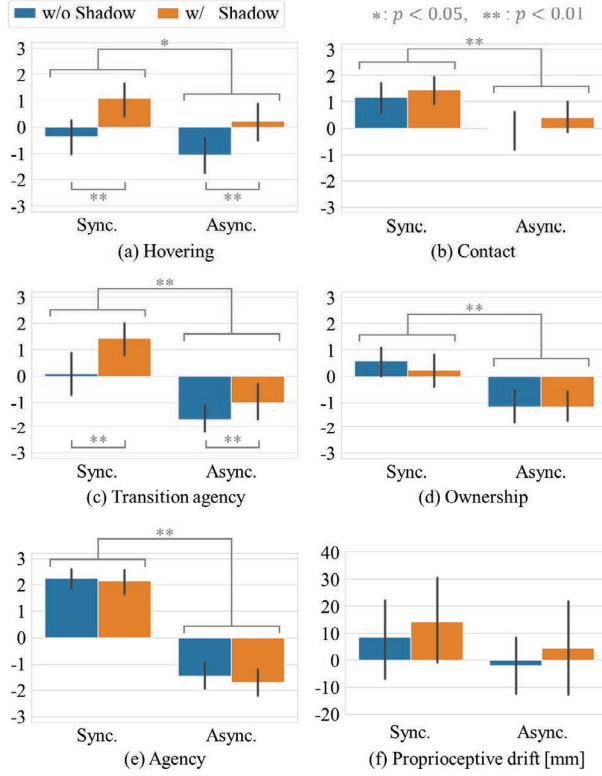


Figure 3: Experimental results.

0.080). However a significant difference was found in the main effect of Synchronization ($F(1, 23) = 32.40, p < 0.001, \eta_p^2 = 0.585$).

For Transition, no significant difference was observed in the interaction effect ($F(1, 23) = 1.60, p = 0.219, \eta_p^2 = 0.065$). However, significant differences were found in the main effects of both Shadow Effect ($F(1, 23) = 10.42, p = 0.004, \eta_p^2 = 0.312$) and Synchronization ($F(1, 23) = 32.67, p < 0.001, \eta_p^2 = 0.587$).

For Ownership, no significant differences were observed in the interaction effect ($F(1, 23) = 0.49, p = 0.492, \eta_p^2 = 0.021$) or the main effect of Shadow Effect ($F(1, 23) = 1.00, p = 0.329, \eta_p^2 = 0.042$). However a significant difference was found in the main effect of Synchronization ($F(1, 23) = 51.92, p < 0.001, \eta_p^2 = 0.693$).

For Agency, no significant differences were observed in the interaction effect ($F(1, 23) = 0.56, p = 0.462, \eta_p^2 = 0.024$) or the main effect of Shadow Effect ($F(1, 23) = 1.10, p = 0.305, \eta_p^2 = 0.046$). However a significant difference was found in the main effect of Synchronization ($F(1, 23) = 139.00, p < 0.001, \eta_p^2 = 0.858$).

For proprioceptive drift, normality was confirmed for each condition. Therefore, a two-way repeated measures ANOVA was conducted. The results did not reveal significant differences in either the main effects (Shadow: $F(1, 23) = 0.67, p = 0.421, \eta_p^2 = 0.028$, Synchronization: $F(1, 23) = 2.24, p = 0.148, \eta_p^2 = 0.089$) or their interaction effect ($F(1, 23) < 0.01, p = 0.972, \eta_p^2 < 0.001$).

3.5 Discussion

The experimental results suggest that the proposed shadow effect significantly enhanced participants' perception of the hovering sensation of the projected hand (Hovering) compared to the conventional condition without the shadow effect. In contrast, participants'

perception of the projected hand being in contact with objects (Contact) scored higher than one on average, regardless of the presence or absence of the shadow effect. This outcome is considered reasonable, as without the shadow effect, the projected hand appears to adhere to the surface of physical objects, as mentioned in Section 1. Moreover, the scores under the w/ Shadow and Sync condition in Fig. 3(e) indicate that participants felt they could control the switching between hovering and contact states themselves. Based on these findings, the proposed shadow effect, which dynamically responds to users' touch inputs, enables users to freely transition the projected hand between hovering and contact states, overcoming the previous limitation of only being able to represent the contact state.

Our group has previously proposed methods that convey the tactile sensation of objects touched by projected hands through haptic feedback [20, 23] and visual [18, 15, 16] or auditory feedback [17] based on pseudo-haptics [5]. These methods have so far always provided tactile feedback for the objects that the projected hand touches, regardless of users' interest in the objects. With this new method, users can express their interest or lack thereof in an object by switching between hovering and contact states, allowing tactile feedback to be provided only for objects of interest.

Even in the Async condition, the presence of the shadow effect improved the perception of hovering compared to when the shadow effect was absent. This implies that not only the operator of the projected hand but also surrounding non-operator users could potentially discern whether the projected hand is hovering or in contact with an object's surface. If this holds, it could help implicitly convey the operator's intent, such as whether they are interested in an object when the projected hand appears near it. Future studies will need to investigate these possibilities in greater detail.

On the other hand, no significant differences were observed in measures such as ownership, agency, or proprioceptive drift with or without the shadow effect. It is well-known that the appearance of the hand resembling a real hand is one of the key factors in eliciting a sense of ownership [6, 21]. In this context, we thought that adding a shadow to the projected hand would make it appear closer to a real hand, thereby enhancing the sense of ownership, but the results did not support this. One possible explanation is that the shadow was not as noticeable compared to the projected hand. In particular, in this experiment, the main task itself encouraged participants to focus more on the fingertips of the projected hand than on the shadow itself. In the future, it will be necessary to examine whether these measures are influenced by using alternative experimental tasks or by enhancing the shadow's presence, such as by altering its color [1, 11].

4 CONCLUSION AND FUTURE WORK

In this study, we proposed a method to simulate the shadow of a projected hand and dynamically adjust its appearance based on the user's touch input. The results of the user study suggested that the proposed method allows users to perceive the projected hand as hovering and enables them to flexibly express both the hovering and contact states of the projected hand. This indicates the potential for users to experience a wider range of interactions with the projected hand. In the future, we plan to conduct detailed investigations that will focus not only on the operator but also include non-operator users. These studies will explore aspects such as shadow color, the distance from users to the projected hand, and the relationship between actual lighting and shadow direction.

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